Name → Diya Goel Roll No. → 2401010179

Topic _____ Date _____

DBMS → Database Management Systems

ASSIGNMENT - 1

Section-A [Introduction to DBMS]

Ques-1 What is a DBMS ? Why is it required ?

A DBMS is software that enables users to define, create, maintain and control access to databases.

- Why it is required :
  - Stores and manages large amounts of data efficiently.
  - Reduces data redundancy
  - Maintains data consistency.
  - Provides data security
  - Supports concurrent access.
  - Provides backup and recovery.

Ques-2 Why is DBMS better than the traditional file system?

DBMS is better because :

- Reduces data redundancy
- Avoids data ~~security~~ inconsistency
- Provides data security
- Supports multi-user access (concurrency control)
- Ensures data integrity
- Provides backup and recovery mechanisms.

In traditional file systems, data duplication and inconsistency are common problems.

Topic _____ Date _____

Ques-3 What is Data Abstraction? Explain its levels.

Data Abstraction means hiding the complex internal details of data storage and showing only necessary information to users.

- Three levels of Data Abstraction:

① Physical level → Describes how data is stored in memory (Files, indexes)

② Logical level → Describes what data is stored and relationship among data.

③ View level → Describes how users view the data (user-specific views)

Ques-4 What is the Three-Schema Architecture? Why is it important?

- The Three-Schema Architecture consists of:

① Internal Schema → Physical storage structure

② Conceptual Schema → Logical structure of database

③ External Schema → User views

- Importance:
  - Provides data independence
  - Separates user applications from physical storage
  - Improves flexibility and security.

Ques-5 What is Data Independence? Why is it needed?

Data independence is the ability to modify schema at one level without affecting other levels.

- Types:

① Physical Data Independence: Changes in physical level do

Topic _____ Date _____

not affect logical level.

② Logical Data Independence → Changes in logical level do not affect external views.

- Need:
  - Easy modification
  - System flexibility
  - Reduced application maintenance.

Ques-6 What is an Entity-Relationship (ER) Model?

The ER Model is a conceptual data model used to represent real-world objects and their relationships graphically.

- Components:
  - Entity
  - Attributes
  - Relationships
  - Primary key

Ques-7 What are Integrity constraints? Why are they important?

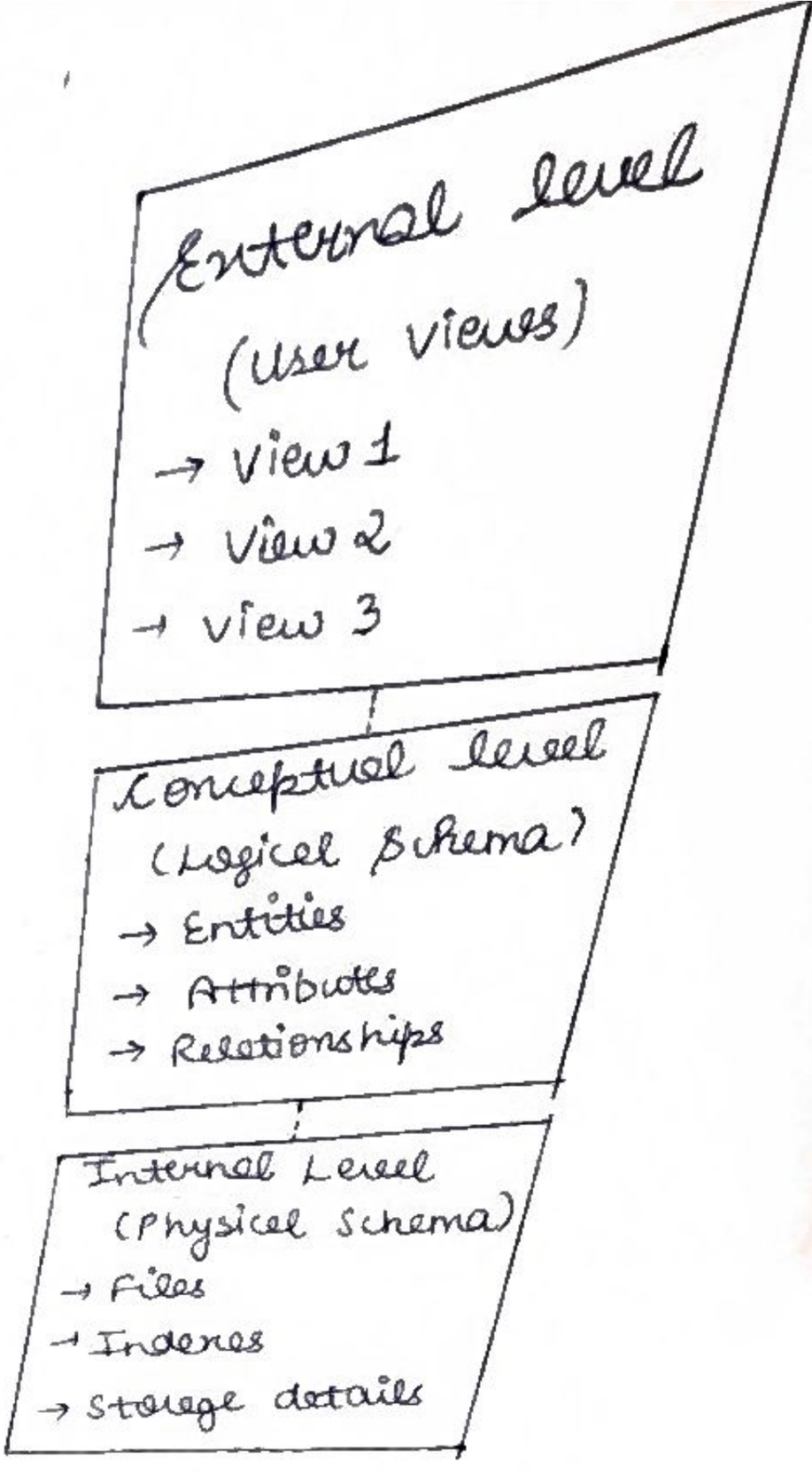
Integrity constraints are rules that ensure accuracy and consistency of data in a database.

- Types:

① Primary Key Constraint → Uniquely identifies records
② Foreign Key Constraint → Maintains referential integrity
③ Unique Constraint → No duplicate values
④ Not Null Constraint → Prevents null values
⑤ Check Constraint → Ensures condition is satisfied

- Importance:
  - Maintains data accuracy
  - Prevents invalid data entry

Ans-1



Ans-2 Entities and Attributes:

Book → (Book-ID, Title, Author, Publisher, Price)

Member → (Member-ID, Name, Address, Phone)

Issue → (Issue-ID, Issue-Date, Return-Date)

Relationships:

- Member borrows Book
- Issue links Member and Book

Primary keys:

- Book-ID, Member-ID, Issue-ID
- One member can borrow many books.

Topic _____ Date _____

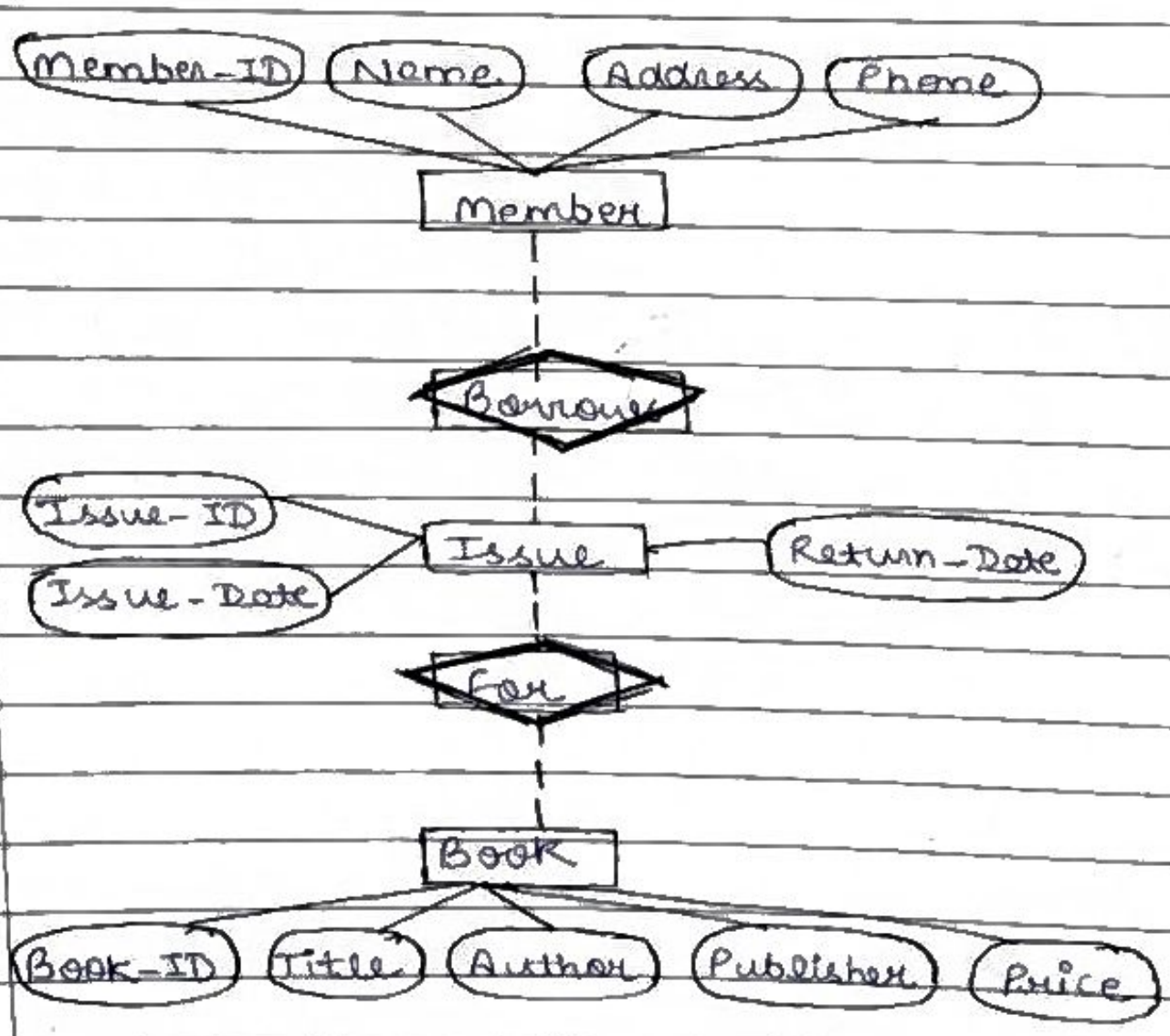
Section-B Design, Draw & Defend (Diagram Based)

1 Draw the three-schema architecture and explain each level.

Internal level → Physical storage details
Conceptual level → Logical database structure
External level → User-specific views

- Benefit → Provides data independence and security.

Ques-2 Draw an ER diagram for a Library Management System.



Ans-3 Entities and Attributes:

Student → Student-ID, Name, Department

Course → Course-ID, Course-Name, Credit

Relationship:

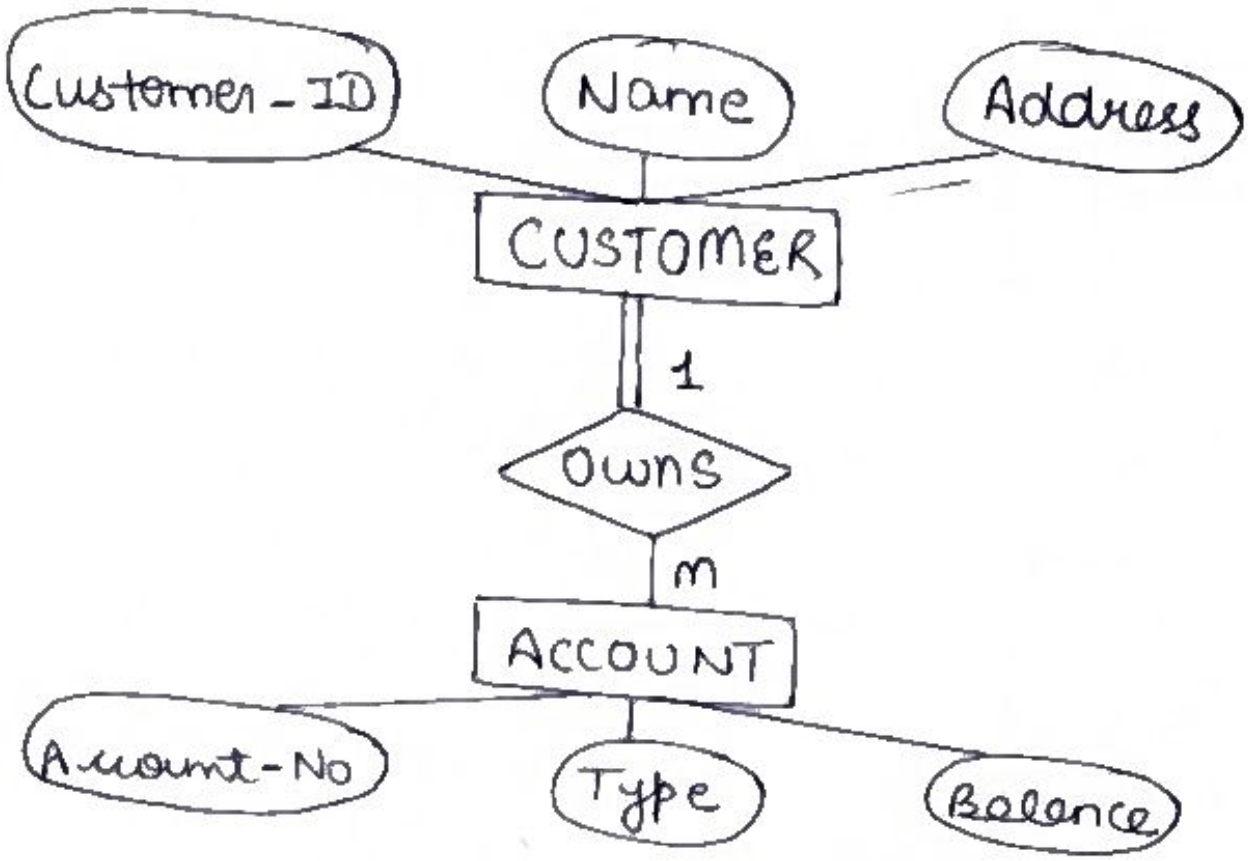
Registers → (Many-to-many relationship)

- One student can register for many courses
- One course can have many students

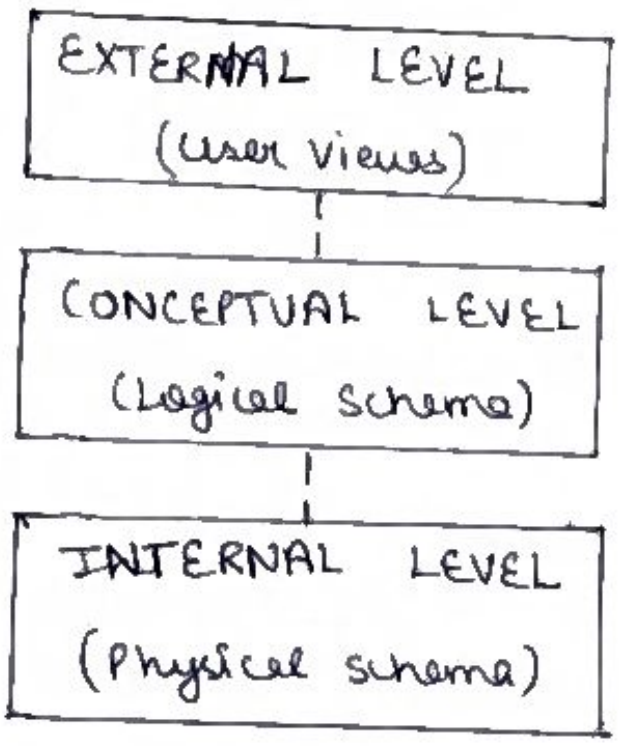
Use a relationship table:

Registration → (Student-ID, Course-ID, Semester)

Ans-1


Customer_ID
Name
Address
CUSTOMER
1
Owns
m
ACCOUNT
Account-No
Type
Balance

Ans-2


EXTERNAL LEVEL
(User Views)
CONCEPTUAL LEVEL
(Logical Schema)
INTERNAL LEVEL
(Physical Schema)

Topic _____ Date _____

- Constraints → ① Patients_ID and Doctor_ID are primary keys.
  ② Appointment references both (foreign keys)
  ③ Data cannot be null

Ques-2 Online shopping Platform →

- Entities: ① Customer (Customer_ID)
  ② Product (Product_ID)
  ③ Order (Order_ID)

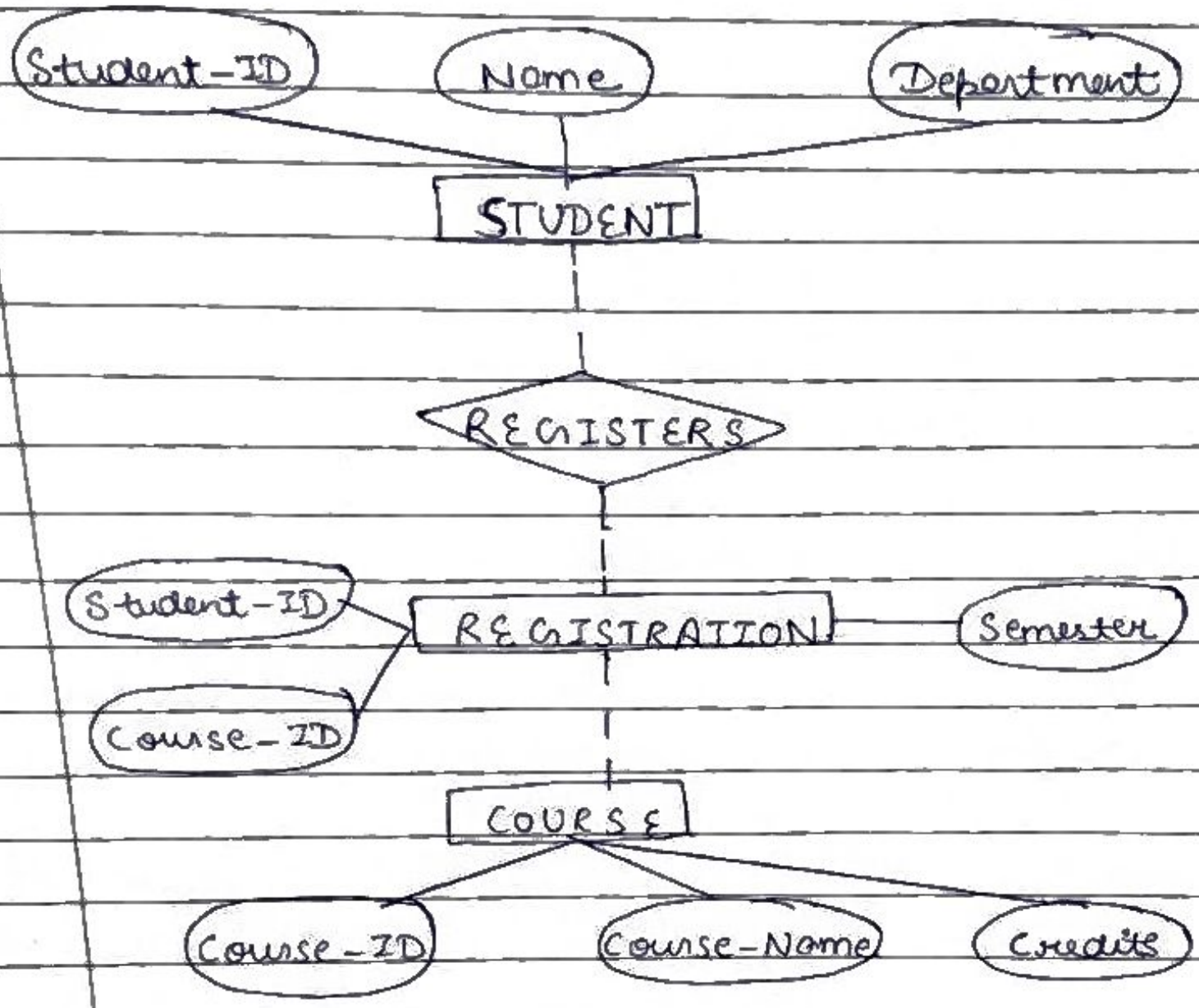
- Keys: ① Primary keys: Customer_ID, Product_ID, Order_ID
  ② Foreign key: Order references Customer_ID.
  ③ Order-Product table for many-to-many.

- Constraints: ① Product price > 0 (check)
  ② Customer email unique
  ③ Order date not null

Ques-3 University stores Student Records - Data Abstraction
① Physical level: Data stored in files and indexes
② Logical level: Tables like Student, Course, Marks
③ View level: · Admin view (all data)
· Teacher view (marks only)
· Student view (own result only)
This ensures security and simplicity.

Topic _____ Date _____

Draw an ER diagram for a student-course registration system.



SECTION-C [Case-Based Analytical Questions]

Ques-1 Hospital Database System →

- Entities : ① Patient (Patient-ID, Name, Age, Contact)
  ② Doctor (Doctor-ID, Name, Specialization)
  ③ Appointment (Appointment-ID, Date, Time)
- Relationships : ① Patient books Appointment
  ② Doctor attends Appointment